FairChild Institute

Code the future Institute

AI-Powered Web Design for Teens

FairChild Institute

Month 1

Computing, Web and Digital Creation

Duration	4 weeks

Month Journey

Week 1: Discover & understand — computer components

Session	Learning	Practical activity	Learner responsibility
1. Concept and inquiry	Explain computer components; define key terms, real-world use cases, common misconceptions and safe practice.	Investigate a familiar example of computer components; draw a diagram, annotate a worked example and discuss trade-offs.	Write a one-page concept note with two examples and three questions about computer components.
2. Mentor demonstration	Observe an instructor-led demonstration of computer components; connect decisions to the monthly product brief.	Follow a short worked example, then modify at least two elements without copying instructions.	Recreate the technique independently and explain how it contributes to youth organisation multi-page website.
3. Guided studio and challenge	Apply computer components in a realistic build; test assumptions, handle errors and use peer feedback.	Implement a small feature or artefact for youth organisation multi-page website; test a normal case and an edge case.	Submit working evidence, a test checklist and a short explanation of design or code choices.
4. Independent build, review and reflection	Synthesise computer components; diagnose problems, critique quality and document improvements.	Complete an independent mini-challenge, review peer work and revise the monthly product.	Submit before/after evidence, a learning reflection and one specific next-step improvement.

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Week 2: Build core skills — files and operating systems

Session	Learning	Practical activity	Learner responsibility
1. Concept and inquiry	Explain files and operating systems; define key terms, real-world use cases, common misconceptions and safe practice.	Investigate a familiar example of files and operating systems; draw a diagram, annotate a worked example and discuss trade-offs.	Write a one-page concept note with two examples and three questions about files and operating systems.
2. Mentor demonstration	Observe an instructor-led demonstration of files and operating systems; connect decisions to the monthly product brief.	Follow a short worked example, then modify at least two elements without copying instructions.	Recreate the technique independently and explain how it contributes to youth organisation multi-page website.
3. Guided studio and challenge	Apply files and operating systems in a realistic build; test assumptions, handle errors and use peer feedback.	Implement a small feature or artefact for youth organisation multi-page website; test a normal case and an edge case.	Submit working evidence, a test checklist and a short explanation of design or code choices.
4. Independent build, review and reflection	Synthesise files and operating systems; diagnose problems, critique quality and document improvements.	Complete an independent mini-challenge, review peer work and revise the monthly product.	Submit before/after evidence, a learning reflection and one specific next-step improvement.

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Week 3: Apply & challenge — web fundamentals

Session	Learning	Practical activity	Learner responsibility
1. Concept and inquiry	Explain web fundamentals; define key terms, real-world use cases, common misconceptions and safe practice.	Investigate a familiar example of web fundamentals; draw a diagram, annotate a worked example and discuss trade-offs.	Write a one-page concept note with two examples and three questions about web fundamentals.
2. Mentor demonstration	Observe an instructor-led demonstration of web fundamentals; connect decisions to the monthly product brief.	Follow a short worked example, then modify at least two elements without copying instructions.	Recreate the technique independently and explain how it contributes to youth organisation multi-page website.
3. Guided studio and challenge	Apply web fundamentals in a realistic build; test assumptions, handle errors and use peer feedback.	Implement a small feature or artefact for youth organisation multi-page website; test a normal case and an edge case.	Submit working evidence, a test checklist and a short explanation of design or code choices.
4. Independent build, review and reflection	Synthesise web fundamentals; diagnose problems, critique quality and document improvements.	Complete an independent mini-challenge, review peer work and revise the monthly product.	Submit before/after evidence, a learning reflection and one specific next-step improvement.

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Week 4: Integrate, assess & reflect — semantic HTML

Session	Learning	Practical activity	Learner responsibility
1. Concept and inquiry	Explain semantic HTML; define key terms, real-world use cases, common misconceptions and safe practice.	Investigate a familiar example of semantic HTML; draw a diagram, annotate a worked example and discuss trade-offs.	Write a one-page concept note with two examples and three questions about semantic HTML.
2. Mentor demonstration	Observe an instructor-led demonstration of semantic HTML; connect decisions to the monthly product brief.	Follow a short worked example, then modify at least two elements without copying instructions.	Recreate the technique independently and explain how it contributes to youth organisation multi-page website.
3. Guided studio and challenge	Apply semantic HTML in a realistic build; test assumptions, handle errors and use peer feedback.	Implement a small feature or artefact for youth organisation multi-page website; test a normal case and an edge case.	Submit working evidence, a test checklist and a short explanation of design or code choices.
4. Independent build, review and reflection	Synthesise semantic HTML; diagnose problems, critique quality and document improvements.	Complete an independent mini-challenge, review peer work and revise the monthly product.	Submit before/after evidence, a learning reflection and one specific next-step improvement.

FairChild Institute

Month 2

CSS, Visual Design and Responsive Websites

Duration 4 weeks

Month Journey

Week 1: Discover & understand — CSS selectors and cascade

Session	Learning	Practical activity	Learner responsibility
1. Concept and inquiry	Explain CSS selectors and cascade; define key terms, real-world use cases, common misconceptions and safe practice.	Investigate a familiar example of CSS selectors and cascade; draw a diagram, annotate a worked example and discuss trade-offs.	Write a one-page concept note with two examples and three questions about CSS selectors and cascade.
2. Mentor demonstration	Observe an instructor-led demonstration of CSS selectors and cascade; connect decisions to the monthly product brief.	Follow a short worked example, then modify at least two elements without copying instructions.	Recreate the technique independently and explain how it contributes to responsive redesign of month 1 website.
3. Guided studio and challenge	Apply CSS selectors and cascade in a realistic build; test assumptions, handle errors and use peer feedback.	Implement a small feature or artefact for responsive redesign of month 1 website; test a normal case and an edge case.	Submit working evidence, a test checklist and a short explanation of design or code choices.
4. Independent build, review and reflection	Synthesise CSS selectors and cascade; diagnose problems, critique quality and document improvements.	Complete an independent mini-challenge, review peer work and revise the monthly product.	Submit before/after evidence, a learning reflection and one specific next-step improvement.

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Week 2: Build core skills — layout and spacing

Session	Learning	Practical activity	Learner responsibility
1. Concept and inquiry	Explain layout and spacing; define key terms, real-world use cases, common misconceptions and safe practice.	Investigate a familiar example of layout and spacing; draw a diagram, annotate a worked example and discuss trade-offs.	Write a one-page concept note with two examples and three questions about layout and spacing.
2. Mentor demonstration	Observe an instructor-led demonstration of layout and spacing; connect decisions to the monthly product brief.	Follow a short worked example, then modify at least two elements without copying instructions.	Recreate the technique independently and explain how it contributes to responsive redesign of month 1 website.
3. Guided studio and challenge	Apply layout and spacing in a realistic build; test assumptions, handle errors and use peer feedback.	Implement a small feature or artefact for responsive redesign of month 1 website; test a normal case and an edge case.	Submit working evidence, a test checklist and a short explanation of design or code choices.
4. Independent build, review and reflection	Synthesise layout and spacing; diagnose problems, critique quality and document improvements.	Complete an independent mini-challenge, review peer work and revise the monthly product.	Submit before/after evidence, a learning reflection and one specific next-step improvement.

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Week 3: Apply & challenge — responsive design

Session	Learning	Practical activity	Learner responsibility
1. Concept and inquiry	Explain responsive design; define key terms, real-world use cases, common misconceptions and safe practice.	Investigate a familiar example of responsive design; draw a diagram, annotate a worked example and discuss trade-offs.	Write a one-page concept note with two examples and three questions about responsive design.
2. Mentor demonstration	Observe an instructor-led demonstration of responsive design; connect decisions to the monthly product brief.	Follow a short worked example, then modify at least two elements without copying instructions.	Recreate the technique independently and explain how it contributes to responsive redesign of month 1 website.
3. Guided studio and challenge	Apply responsive design in a realistic build; test assumptions, handle errors and use peer feedback.	Implement a small feature or artefact for responsive redesign of month 1 website; test a normal case and an edge case.	Submit working evidence, a test checklist and a short explanation of design or code choices.
4. Independent build, review and reflection	Synthesise responsive design; diagnose problems, critique quality and document improvements.	Complete an independent mini-challenge, review peer work and revise the monthly product.	Submit before/after evidence, a learning reflection and one specific next-step improvement.

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Week 4: Integrate, assess & reflect — accessible web forms

Session	Learning	Practical activity	Learner responsibility
1. Concept and inquiry	Explain accessible web forms; define key terms, real-world use cases, common misconceptions and safe practice.	Investigate a familiar example of accessible web forms; draw a diagram, annotate a worked example and discuss trade-offs.	Write a one-page concept note with two examples and three questions about accessible web forms.
2. Mentor demonstration	Observe an instructor-led demonstration of accessible web forms; connect decisions to the monthly product brief.	Follow a short worked example, then modify at least two elements without copying instructions.	Recreate the technique independently and explain how it contributes to responsive redesign of month 1 website.
3. Guided studio and challenge	Apply accessible web forms in a realistic build; test assumptions, handle errors and use peer feedback.	Implement a small feature or artefact for responsive redesign of month 1 website; test a normal case and an edge case.	Submit working evidence, a test checklist and a short explanation of design or code choices.
4. Independent build, review and reflection	Synthesise accessible web forms; diagnose problems, critique quality and document improvements.	Complete an independent mini-challenge, review peer work and revise the monthly product.	Submit before/after evidence, a learning reflection and one specific next-step improvement.

FairChild Institute

Month 3

JavaScript and Interactive Applications

Duration	4 weeks

Month Journey

Week 1: Discover & understand — variables and logic

Session	Learning	Practical activity	Learner responsibility
1. Concept and inquiry	Explain variables and logic; define key terms, real-world use cases, common misconceptions and safe practice.	Investigate a familiar example of variables and logic; draw a diagram, annotate a worked example and discuss trade-offs.	Write a one-page concept note with two examples and three questions about variables and logic.
2. Mentor demonstration	Observe an instructor-led demonstration of variables and logic; connect decisions to the monthly product brief.	Follow a short worked example, then modify at least two elements without copying instructions.	Recreate the technique independently and explain how it contributes to interactive youth organisation application.
3. Guided studio and challenge	Apply variables and logic in a realistic build; test assumptions, handle errors and use peer feedback.	Implement a small feature or artefact for interactive youth organisation application; test a normal case and an edge case.	Submit working evidence, a test checklist and a short explanation of design or code choices.
4. Independent build, review and reflection	Synthesise variables and logic; diagnose problems, critique quality and document improvements.	Complete an independent mini-challenge, review peer work and revise the monthly product.	Submit before/after evidence, a learning reflection and one specific next-step improvement.

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Week 2: Build core skills — DOM and events

Session	Learning	Practical activity	Learner responsibility
1. Concept and inquiry	Explain DOM and events; define key terms, real-world use cases, common misconceptions and safe practice.	Investigate a familiar example of DOM and events; draw a diagram, annotate a worked example and discuss trade-offs.	Write a one-page concept note with two examples and three questions about DOM and events.
2. Mentor demonstration	Observe an instructor-led demonstration of DOM and events; connect decisions to the monthly product brief.	Follow a short worked example, then modify at least two elements without copying instructions.	Recreate the technique independently and explain how it contributes to interactive youth organisation application.
3. Guided studio and challenge	Apply DOM and events in a realistic build; test assumptions, handle errors and use peer feedback.	Implement a small feature or artefact for interactive youth organisation application; test a normal case and an edge case.	Submit working evidence, a test checklist and a short explanation of design or code choices.
4. Independent build, review and reflection	Synthesise DOM and events; diagnose problems, critique quality and document improvements.	Complete an independent mini-challenge, review peer work and revise the monthly product.	Submit before/after evidence, a learning reflection and one specific next-step improvement.

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Week 3: Apply & challenge — arrays and objects

Session	Learning	Practical activity	Learner responsibility
1. Concept and inquiry	Explain arrays and objects; define key terms, real-world use cases, common misconceptions and safe practice.	Investigate a familiar example of arrays and objects; draw a diagram, annotate a worked example and discuss trade-offs.	Write a one-page concept note with two examples and three questions about arrays and objects.
2. Mentor demonstration	Observe an instructor-led demonstration of arrays and objects; connect decisions to the monthly product brief.	Follow a short worked example, then modify at least two elements without copying instructions.	Recreate the technique independently and explain how it contributes to interactive youth organisation application.
3. Guided studio and challenge	Apply arrays and objects in a realistic build; test assumptions, handle errors and use peer feedback.	Implement a small feature or artefact for interactive youth organisation application; test a normal case and an edge case.	Submit working evidence, a test checklist and a short explanation of design or code choices.
4. Independent build, review and reflection	Synthesise arrays and objects; diagnose problems, critique quality and document improvements.	Complete an independent mini-challenge, review peer work and revise the monthly product.	Submit before/after evidence, a learning reflection and one specific next-step improvement.

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Week 4: Integrate, assess & reflect — validation and debugging

Session	Learning	Practical activity	Learner responsibility
1. Concept and inquiry	Explain validation and debugging; define key terms, real-world use cases, common misconceptions and safe practice.	Investigate a familiar example of validation and debugging; draw a diagram, annotate a worked example and discuss trade-offs.	Write a one-page concept note with two examples and three questions about validation and debugging.
2. Mentor demonstration	Observe an instructor-led demonstration of validation and debugging; connect decisions to the monthly product brief.	Follow a short worked example, then modify at least two elements without copying instructions.	Recreate the technique independently and explain how it contributes to interactive youth organisation application.
3. Guided studio and challenge	Apply validation and debugging in a realistic build; test assumptions, handle errors and use peer feedback.	Implement a small feature or artefact for interactive youth organisation application; test a normal case and an edge case.	Submit working evidence, a test checklist and a short explanation of design or code choices.
4. Independent build, review and reflection	Synthesise validation and debugging; diagnose problems, critique quality and document improvements.	Complete an independent mini-challenge, review peer work and revise the monthly product.	Submit before/after evidence, a learning reflection and one specific next-step improvement.

FairChild Institute

Month 4

Programming Foundations and Problem Solving

Duration 4 weeks

Month Journey

Week 1: Discover & understand — algorithms and pseudocode

Session	Learning	Practical activity	Learner responsibility
1. Concept and inquiry	Explain algorithms and pseudocode; define key terms, real-world use cases, common misconceptions and safe practice.	Investigate a familiar example of algorithms and pseudocode; draw a diagram, annotate a worked example and discuss trade-offs.	Write a one-page concept note with two examples and three questions about algorithms and pseudocode.
2. Mentor demonstration	Observe an instructor-led demonstration of algorithms and pseudocode; connect decisions to the monthly product brief.	Follow a short worked example, then modify at least two elements without copying instructions.	Recreate the technique independently and explain how it contributes to console-based problem-solving toolkit.
3. Guided studio and challenge	Apply algorithms and pseudocode in a realistic build; test assumptions, handle errors and use peer feedback.	Implement a small feature or artefact for console-based problem-solving toolkit; test a normal case and an edge case.	Submit working evidence, a test checklist and a short explanation of design or code choices.
4. Independent build, review and reflection	Synthesise algorithms and pseudocode; diagnose problems, critique quality and document improvements.	Complete an independent mini-challenge, review peer work and revise the monthly product.	Submit before/after evidence, a learning reflection and one specific next-step improvement.

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Week 2: Build core skills — functions and decomposition

Session	Learning	Practical activity	Learner responsibility
1. Concept and inquiry	Explain functions and decomposition; define key terms, real-world use cases, common misconceptions and safe practice.	Investigate a familiar example of functions and decomposition; draw a diagram, annotate a worked example and discuss trade-offs.	Write a one-page concept note with two examples and three questions about functions and decomposition.
2. Mentor demonstration	Observe an instructor-led demonstration of functions and decomposition; connect decisions to the monthly product brief.	Follow a short worked example, then modify at least two elements without copying instructions.	Recreate the technique independently and explain how it contributes to console-based problem-solving toolkit.
3. Guided studio and challenge	Apply functions and decomposition in a realistic build; test assumptions, handle errors and use peer feedback.	Implement a small feature or artefact for console-based problem-solving toolkit; test a normal case and an edge case.	Submit working evidence, a test checklist and a short explanation of design or code choices.
4. Independent build, review and reflection	Synthesise functions and decomposition; diagnose problems, critique quality and document improvements.	Complete an independent mini-challenge, review peer work and revise the monthly product.	Submit before/after evidence, a learning reflection and one specific next-step improvement.

FairChild Institute

Week 3: Apply & challenge — loops and branching

Session	Learning	Practical activity	Learner responsibility
1. Concept and inquiry	Explain loops and branching; define key terms, real-world use cases, common misconceptions and safe practice.	Investigate a familiar example of loops and branching; draw a diagram, annotate a worked example and discuss trade-offs.	Write a one-page concept note with two examples and three questions about loops and branching.
2. Mentor demonstration	Observe an instructor-led demonstration of loops and branching; connect decisions to the monthly product brief.	Follow a short worked example, then modify at least two elements without copying instructions.	Recreate the technique independently and explain how it contributes to console-based problem-solving toolkit.
3. Guided studio and challenge	Apply loops and branching in a realistic build; test assumptions, handle errors and use peer feedback.	Implement a small feature or artefact for console-based problem-solving toolkit; test a normal case and an edge case.	Submit working evidence, a test checklist and a short explanation of design or code choices.
4. Independent build, review and reflection	Synthesise loops and branching; diagnose problems, critique quality and document improvements.	Complete an independent mini-challenge, review peer work and revise the monthly product.	Submit before/after evidence, a learning reflection and one specific next-step improvement.

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Week 4: Integrate, assess & reflect — debugging methods

Session	Learning	Practical activity	Learner responsibility
1. Concept and inquiry	Explain debugging methods; define key terms, real-world use cases, common misconceptions and safe practice.	Investigate a familiar example of debugging methods; draw a diagram, annotate a worked example and discuss trade-offs.	Write a one-page concept note with two examples and three questions about debugging methods.
2. Mentor demonstration	Observe an instructor-led demonstration of debugging methods; connect decisions to the monthly product brief.	Follow a short worked example, then modify at least two elements without copying instructions.	Recreate the technique independently and explain how it contributes to console-based problem-solving toolkit.
3. Guided studio and challenge	Apply debugging methods in a realistic build; test assumptions, handle errors and use peer feedback.	Implement a small feature or artefact for console-based problem-solving toolkit; test a normal case and an edge case.	Submit working evidence, a test checklist and a short explanation of design or code choices.
4. Independent build, review and reflection	Synthesise debugging methods; diagnose problems, critique quality and document improvements.	Complete an independent mini-challenge, review peer work and revise the monthly product.	Submit before/after evidence, a learning reflection and one specific next-step improvement.

FairChild Institute

Month 5

Python and Automation

Duration 4 weeks

Month Journey

Week 1: Discover & understand — Python syntax and data types

Session	Learning	Practical activity	Learner responsibility
1. Concept and inquiry	Explain Python syntax and data types; define key terms, real-world use cases, common misconceptions and safe practice.	Investigate a familiar example of Python syntax and data types; draw a diagram, annotate a worked example and discuss trade-offs.	Write a one-page concept note with two examples and three questions about Python syntax and data types.
2. Mentor demonstration	Observe an instructor-led demonstration of Python syntax and data types; connect decisions to the monthly product brief.	Follow a short worked example, then modify at least two elements without copying instructions.	Recreate the technique independently and explain how it contributes to personal productivity automation.
3. Guided studio and challenge	Apply Python syntax and data types in a realistic build; test assumptions, handle errors and use peer feedback.	Implement a small feature or artefact for personal productivity automation; test a normal case and an edge case.	Submit working evidence, a test checklist and a short explanation of design or code choices.
4. Independent build, review and reflection	Synthesise Python syntax and data types; diagnose problems, critique quality and document improvements.	Complete an independent mini-challenge, review peer work and revise the monthly product.	Submit before/after evidence, a learning reflection and one specific next-step improvement.

Week 2: Build core skills — functions and files

Session	Learning	Practical activity	Learner responsibility
1. Concept and inquiry	Explain functions and files; define key terms, real-world use cases, common misconceptions and safe practice.	Investigate a familiar example of functions and files; draw a diagram, annotate a worked example and discuss trade-offs.	Write a one-page concept note with two examples and three questions about functions and decomposition.
2. Mentor demonstration	Observe an instructor-led demonstration of functions and decomposition; connect decisions to the monthly product brief.	Follow a short worked example, then modify at least two elements without copying instructions.	Recreate the technique independently and explain how it contributes to personal productivity automation.
3. Guided studio and challenge	Apply functions and files in a realistic build; test assumptions, handle errors and use peer feedback.	Implement a small feature or artefact for personal productivity automation; test a normal case and an edge case.	Submit working evidence, a test checklist and a short explanation of design or code choices.
4. Independent build, review and reflection	Synthesise functions and files; diagnose problems, critique quality and document improvements.	Complete an independent mini-challenge, review peer work and revise the monthly product.	Submit before/after evidence, a learning reflection and one specific next-step improvement.

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Week 3: Apply & challenge — modules and errors

Session	Learning	Practical activity	Learner responsibility
1. Concept and inquiry	Explain modules and errors; define key terms, real-world use cases, common misconceptions and safe practice.	Investigate a familiar example of modules and errors; draw a diagram, annotate a worked example and discuss trade-offs.	Write a one-page concept note with two examples and three questions about modules and errors.
2. Mentor demonstration	Observe an instructor-led demonstration of modules and errors; connect decisions to the monthly product brief.	Follow a short worked example, then modify at least two elements without copying instructions.	Recreate the technique independently and explain how it contributes to personal productivity automation.
3. Guided studio and challenge	Apply modules and errors in a realistic build; test assumptions, handle errors and use peer feedback.	Implement a small feature or artefact for personal productivity automation; test a normal case and an edge case.	Submit working evidence, a test checklist and a short explanation of design or code choices.
4. Independent build, review and reflection	Synthesise modules and errors; diagnose problems, critique quality and document improvements.	Complete an independent mini-challenge, review peer work and revise the monthly product.	Submit before/after evidence, a learning reflection and one specific next-step improvement.

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Week 4: Integrate, assess & reflect — automation scripts

Session	Learning	Practical activity	Learner responsibility
1. Concept and inquiry	Explain automation scripts; define key terms, real-world use cases, common misconceptions and safe practice.	Investigate a familiar example of automation scripts; draw a diagram, annotate a worked example and discuss trade-offs.	Write a one-page concept note with two examples and three questions about automation scripts.
2. Mentor demonstration	Observe an instructor-led demonstration of automation scripts; connect decisions to the monthly product brief.	Follow a short worked example, then modify at least two elements without copying instructions.	Recreate the technique independently and explain how it contributes to personal productivity automation.
3. Guided studio and challenge	Apply automation scripts in a realistic build; test assumptions, handle errors and use peer feedback.	Implement a small feature or artefact for personal productivity automation; test a normal case and an edge case.	Submit working evidence, a test checklist and a short explanation of design or code choices.
4. Independent build, review and reflection	Synthesise automation scripts; diagnose problems, critique quality and document improvements.	Complete an independent mini-challenge, review peer work and revise the monthly product.	Submit before/after evidence, a learning reflection and one specific next-step improvement.

FairChild Institute

Month 6

Algorithms, Data Structures and Computational Thinking

Duration 4 weeks

Month Journey

Week 1: Discover & understand — lists stacks and queues

Session	Learning	Practical activity	Learner responsibility
1. Concept and inquiry	Explain lists stacks and queues; define key terms, real-world use cases, common misconceptions and safe practice.	Investigate a familiar example of lists stacks and queues; draw a diagram, annotate a worked example and discuss trade-offs.	Write a one-page concept note with two examples and three questions about lists stacks and queues.
2. Mentor demonstration	Observe an instructor-led demonstration of lists stacks and queues; connect decisions to the monthly product brief.	Follow a short worked example, then modify at least two elements without copying instructions.	Recreate the technique independently and explain how it contributes to algorithm challenge portfolio.
3. Guided studio and challenge	Apply lists stacks and queues in a realistic build; test assumptions, handle errors and use peer feedback.	Implement a small feature or artefact for algorithm challenge portfolio; test a normal case and an edge case.	Submit working evidence, a test checklist and a short explanation of design or code choices.
4. Independent build, review and reflection	Synthesise lists stacks and queues; diagnose problems, critique quality and document improvements.	Complete an independent mini-challenge, review peer work and revise the monthly product.	Submit before/after evidence, a learning reflection and one specific next-step improvement.

Week 2: Build core skills — searching and sorting

Session	Learning	Practical activity	Learner responsibility
1. Concept and inquiry	Explain searching and sorting; define key terms, real-world use cases, common misconceptions and safe practice.	Investigate a familiar example of searching and sorting; draw a diagram, annotate a worked example and discuss trade-offs.	Write a one-page concept note with two examples and three questions about searching and sorting.
2. Mentor demonstration	Observe an instructor-led demonstration of searching and sorting; connect decisions to the monthly product brief.	Follow a short worked example, then modify at least two elements without copying instructions.	Recreate the technique independently and explain how it contributes to algorithm challenge portfolio.
3. Guided studio and challenge	Apply searching and sorting in a realistic build; test assumptions, handle errors and use peer feedback.	Implement a small feature or artefact for algorithm challenge portfolio; test a normal case and an edge case.	Submit working evidence, a test checklist and a short explanation of design or code choices.
4. Independent build, review and reflection	Synthesise searching and sorting; diagnose problems, critique quality and document improvements.	Complete an independent mini-challenge, review peer work and revise the monthly product.	Submit before/after evidence, a learning reflection and one specific next-step improvement.

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Week 3: Apply & challenge — complexity intuition

Session	Learning	Practical activity	Learner responsibility
1. Concept and inquiry	Explain complexity intuition; define key terms, real-world use cases, common misconceptions and safe practice.	Investigate a familiar example of complexity intuition; draw a diagram, annotate a worked example and discuss trade-offs.	Write a one-page concept note with two examples and three questions about complexity intuition.
2. Mentor demonstration	Observe an instructor-led demonstration of complexity intuition; connect decisions to the monthly product brief.	Follow a short worked example, then modify at least two elements without copying instructions.	Recreate the technique independently and explain how it contributes to algorithm challenge portfolio.
3. Guided studio and challenge	Apply complexity intuition in a realistic build; test assumptions, handle errors and use peer feedback.	Implement a small feature or artefact for algorithm challenge portfolio; test a normal case and an edge case.	Submit working evidence, a test checklist and a short explanation of design or code choices.
4. Independent build, review and reflection	Synthesise complexity intuition; diagnose problems, critique quality and document improvements.	Complete an independent mini-challenge, review peer work and revise the monthly product.	Submit before/after evidence, a learning reflection and one specific next-step improvement.

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Week 4: Integrate, assess & reflect — problem-solving practice

Session	Learning	Practical activity	Learner responsibility
1. Concept and inquiry	Explain problem-solving practice; define key terms, real-world use cases, common misconceptions and safe practice.	Investigate a familiar example of problem-solving practice; draw a diagram, annotate a worked example and discuss trade-offs.	Write a one-page concept note with two examples and three questions about problem-solving practice.
2. Mentor demonstration	Observe an instructor-led demonstration of problem-solving practice; connect decisions to the monthly product brief.	Follow a short worked example, then modify at least two elements without copying instructions.	Recreate the technique independently and explain how it contributes to algorithm challenge portfolio.
3. Guided studio and challenge	Apply problem-solving practice in a realistic build; test assumptions, handle errors and use peer feedback.	Implement a small feature or artefact for algorithm challenge portfolio; test a normal case and an edge case.	Submit working evidence, a test checklist and a short explanation of design or code choices.
4. Independent build, review and reflection	Synthesise problem-solving practice; diagnose problems, critique quality and document improvements.	Complete an independent mini-challenge, review peer work and revise the monthly product.	Submit before/after evidence, a learning reflection and one specific next-step improvement.

FairChild Institute

Month 7

Backend, HTTP and APIs

Duration 4 weeks

Month Journey

Week 1: Discover & understand — client-server model

Session	Learning	Practical activity	Learner responsibility
1. Concept and inquiry	Explain client-server model; define key terms, real-world use cases, common misconceptions and safe practice.	Investigate a familiar example of client-server model; draw a diagram, annotate a worked example and discuss trade-offs.	Write a one-page concept note with two examples and three questions about client-server model.
2. Mentor demonstration	Observe an instructor-led demonstration of client-server model; connect decisions to the monthly product brief.	Follow a short worked example, then modify at least two elements without copying instructions.	Recreate the technique independently and explain how it contributes to simple backend api.
3. Guided studio and challenge	Apply client-server model in a realistic build; test assumptions, handle errors and use peer feedback.	Implement a small feature or artefact for simple backend api; test a normal case and an edge case.	Submit working evidence, a test checklist and a short explanation of design or code choices.
4. Independent build, review and reflection	Synthesise client-server model; diagnose problems, critique quality and document improvements.	Complete an independent mini-challenge, review peer work and revise the monthly product.	Submit before/after evidence, a learning reflection and one specific next-step improvement.

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Week 2: Build core skills — HTTP methods and status

Session	Learning	Practical activity	Learner responsibility
1. Concept and inquiry	Explain HTTP methods and status; define key terms, real-world use cases, common misconceptions and safe practice.	Investigate a familiar example of HTTP methods and status; draw a diagram, annotate a worked example and discuss trade-offs.	Write a one-page concept note with two examples and three questions about HTTP methods and status.
2. Mentor demonstration	Observe an instructor-led demonstration of HTTP methods and status; connect decisions to the monthly product brief.	Follow a short worked example, then modify at least two elements without copying instructions.	Recreate the technique independently and explain how it contributes to simple backend api.
3. Guided studio and challenge	Apply HTTP methods and status in a realistic build; test assumptions, handle errors and use peer feedback.	Implement a small feature or artefact for simple backend api; test a normal case and an edge case.	Submit working evidence, a test checklist and a short explanation of design or code choices.
4. Independent build, review and reflection	Synthesise HTTP methods and status; diagnose problems, critique quality and document improvements.	Complete an independent mini-challenge, review peer work and revise the monthly product.	Submit before/after evidence, a learning reflection and one specific next-step improvement.

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Week 3: Apply & challenge — REST API basics

Session	Learning	Practical activity	Learner responsibility
1. Concept and inquiry	Explain REST API basics; define key terms, real-world use cases, common misconceptions and safe practice.	Investigate a familiar example of REST API basics; draw a diagram, annotate a worked example and discuss trade-offs.	Write a one-page concept note with two examples and three questions about REST API basics.
2. Mentor demonstration	Observe an instructor-led demonstration of REST API basics; connect decisions to the monthly product brief.	Follow a short worked example, then modify at least two elements without copying instructions.	Recreate the technique independently and explain how it contributes to simple backend api.
3. Guided studio and challenge	Apply REST API basics in a realistic build; test assumptions, handle errors and use peer feedback.	Implement a small feature or artefact for simple backend api; test a normal case and an edge case.	Submit working evidence, a test checklist and a short explanation of design or code choices.
4. Independent build, review and reflection	Synthesise REST API basics; diagnose problems, critique quality and document improvements.	Complete an independent mini-challenge, review peer work and revise the monthly product.	Submit before/after evidence, a learning reflection and one specific next-step improvement.

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Week 4: Integrate, assess & reflect — API authentication and errors

Session	Learning	Practical activity	Learner responsibility
1. Concept and inquiry	Explain API authentication and errors; define key terms, real-world use cases, common misconceptions and safe practice.	Investigate a familiar example of API authentication and errors; draw a diagram, annotate a worked example and discuss trade-offs.	Write a one-page concept note with two examples and three questions about API authentication and errors.
2. Mentor demonstration	Observe an instructor-led demonstration of API authentication and errors; connect decisions to the monthly product brief.	Follow a short worked example, then modify at least two elements without copying instructions.	Recreate the technique independently and explain how it contributes to simple backend api.
3. Guided studio and challenge	Apply API authentication and errors in a realistic build; test assumptions, handle errors and use peer feedback.	Implement a small feature or artefact for simple backend api; test a normal case and an edge case.	Submit working evidence, a test checklist and a short explanation of design or code choices.
4. Independent build, review and reflection	Synthesise API authentication and errors; diagnose problems, critique quality and document improvements.	Complete an independent mini-challenge, review peer work and revise the monthly product.	Submit before/after evidence, a learning reflection and one specific next-step improvement.

FairChild Institute

Month 8

PostgreSQL, SQL and Data Modelling

Duration	4 weeks

Month Journey

Week 1: Discover & understand — relational models

Session	Learning	Practical activity	Learner responsibility
1. Concept and inquiry	Explain relational models; define key terms, real-world use cases, common misconceptions and safe practice.	Investigate a familiar example of relational models; draw a diagram, annotate a worked example and discuss trade-offs.	Write a one-page concept note with two examples and three questions about relational models.
2. Mentor demonstration	Observe an instructor-led demonstration of relational models; connect decisions to the monthly product brief.	Follow a short worked example, then modify at least two elements without copying instructions.	Recreate the technique independently and explain how it contributes to community resource database.
3. Guided studio and challenge	Apply relational models in a realistic build; test assumptions, handle errors and use peer feedback.	Implement a small feature or artefact for community resource database; test a normal case and an edge case.	Submit working evidence, a test checklist and a short explanation of design or code choices.
4. Independent build, review and reflection	Synthesise relational models; diagnose problems, critique quality and document improvements.	Complete an independent mini-challenge, review peer work and revise the monthly product.	Submit before/after evidence, a learning reflection and one specific next-step improvement.

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Week 2: Build core skills — SQL queries

Session	Learning	Practical activity	Learner responsibility
1. Concept and inquiry	Explain SQL queries; define key terms, real-world use cases, common misconceptions and safe practice.	Investigate a familiar example of SQL queries; draw a diagram, annotate a worked example and discuss trade-offs.	Write a one-page concept note with two examples and three questions about SQL queries.
2. Mentor demonstration	Observe an instructor-led demonstration of SQL queries; connect decisions to the monthly product brief.	Follow a short worked example, then modify at least two elements without copying instructions.	Recreate the technique independently and explain how it contributes to community resource database.
3. Guided studio and challenge	Apply SQL queries in a realistic build; test assumptions, handle errors and use peer feedback.	Implement a small feature or artefact for community resource database; test a normal case and an edge case.	Submit working evidence, a test checklist and a short explanation of design or code choices.
4. Independent build, review and reflection	Synthesise SQL queries; diagnose problems, critique quality and document improvements.	Complete an independent mini-challenge, review peer work and revise the monthly product.	Submit before/after evidence, a learning reflection and one specific next-step improvement.

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Week 3: Apply & challenge — joins and aggregates

Session	Learning	Practical activity	Learner responsibility
1. Concept and inquiry	Explain joins and aggregates; define key terms, real-world use cases, common misconceptions and safe practice.	Investigate a familiar example of joins and aggregates; draw a diagram, annotate a worked example and discuss trade-offs.	Write a one-page concept note with two examples and three questions about joins and aggregates.
2. Mentor demonstration	Observe an instructor-led demonstration of joins and aggregates; connect decisions to the monthly product brief.	Follow a short worked example, then modify at least two elements without copying instructions.	Recreate the technique independently and explain how it contributes to community resource database.
3. Guided studio and challenge	Apply joins and aggregates in a realistic build; test assumptions, handle errors and use peer feedback.	Implement a small feature or artefact for community resource database; test a normal case and an edge case.	Submit working evidence, a test checklist and a short explanation of design or code choices.
4. Independent build, review and reflection	Synthesise joins and aggregates; diagnose problems, critique quality and document improvements.	Complete an independent mini-challenge, review peer work and revise the monthly product.	Submit before/after evidence, a learning reflection and one specific next-step improvement.

Week 4: Integrate, assess & reflect — schema constraints and migrations

Session	Learning	Practical activity	Learner responsibility
1. Concept and inquiry	Explain schema constraints and migrations; define key terms, real-world use cases, common misconceptions and safe practice.	Investigate a familiar example of schema constraints and migrations; draw a diagram, annotate a worked example and discuss trade-offs.	Write a one-page concept note with two examples and three questions about schema constraints and migrations.
2. Mentor demonstration	Observe an instructor-led demonstration of schema constraints and migrations; connect decisions to the monthly product brief.	Follow a short worked example, then modify at least two elements without copying instructions.	Recreate the technique independently and explain how it contributes to community resource database.
3. Guided studio and challenge	Apply schema constraints and migrations in a realistic build; test assumptions, handle errors and use peer feedback.	Implement a small feature or artefact for community resource database; test a normal case and an edge case.	Submit working evidence, a test checklist and a short explanation of design or code choices.
4. Independent build, review and reflection	Synthesise schema constraints and migrations; diagnose problems, critique quality and document improvements.	Complete an independent mini-challenge, review peer work and revise the monthly product.	Submit before/after evidence, a learning reflection and one specific next-step improvement.

FairChild Institute

Month 9

Full-Stack Product Engineering

Duration	4 weeks

Month Journey

Week 1: Discover & understand — frontend-backend integration

Session	Learning	Practical activity	Learner responsibility
1. Concept and inquiry	Explain frontend-backend integration; define key terms, real-world use cases, common misconceptions and safe practice.	Investigate a familiar example of frontend-backend integration; draw a diagram, annotate a worked example and discuss trade-offs.	Write a one-page concept note with two examples and three questions about frontend-backend integration.
2. Mentor demonstration	Observe an instructor-led demonstration of frontend-backend integration; connect decisions to the monthly product brief.	Follow a short worked example, then modify at least two elements without copying instructions.	Recreate the technique independently and explain how it contributes to full-stack community platform.
3. Guided studio and challenge	Apply frontend-backend integration in a realistic build; test assumptions, handle errors and use peer feedback.	Implement a small feature or artefact for full-stack community platform; test a normal case and an edge case.	Submit working evidence, a test checklist and a short explanation of design or code choices.
4. Independent build, review and reflection	Synthesise frontend-backend integration; diagnose problems, critique quality and document improvements.	Complete an independent mini-challenge, review peer work and revise the monthly product.	Submit before/after evidence, a learning reflection and one specific next-step improvement.

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Week 2: Build core skills — CRUD workflows

Session	Learning	Practical activity	Learner responsibility
1. Concept and inquiry	Explain CRUD workflows; define key terms, real-world use cases, common misconceptions and safe practice.	Investigate a familiar example of CRUD workflows; draw a diagram, annotate a worked example and discuss trade-offs.	Write a one-page concept note with two examples and three questions about CRUD workflows.
2. Mentor demonstration	Observe an instructor-led demonstration of CRUD workflows; connect decisions to the monthly product brief.	Follow a short worked example, then modify at least two elements without copying instructions.	Recreate the technique independently and explain how it contributes to full-stack community platform.
3. Guided studio and challenge	Apply CRUD workflows and status in a realistic build; test assumptions, handle errors and use peer feedback.	Implement a small feature or artefact for full-stack community platform; test a normal case and an edge case.	Submit working evidence, a test checklist and a short explanation of design or code choices.
4. Independent build, review and reflection	Synthesise CRUD workflows; diagnose problems, critique quality and document improvements.	Complete an independent mini-challenge, review peer work and revise the monthly product.	Submit before/after evidence, a learning reflection and one specific next-step improvement.

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Week 3: Apply & challenge — authentication principles

Session	Learning	Practical activity	Learner responsibility
1. Concept and inquiry	Explain authentication principles; define key terms, real-world use cases, common misconceptions and safe practice.	Investigate a familiar example of authentication principles; draw a diagram, annotate a worked example and discuss trade-offs.	Write a one-page concept note with two examples and three questions about authentication principles.
2. Mentor demonstration	Observe an instructor-led demonstration of authentication principles; connect decisions to the monthly product brief.	Follow a short worked example, then modify at least two elements without copying instructions.	Recreate the technique independently and explain how it contributes to full-stack community platform.
3. Guided studio and challenge	Apply authentication principles in a realistic build; test assumptions, handle errors and use peer feedback.	Implement a small feature or artefact for full-stack community platform; test a normal case and an edge case.	Submit working evidence, a test checklist and a short explanation of design or code choices.
4. Independent build, review and reflection	Synthesise authentication principless; diagnose problems, critique quality and document improvements.	Complete an independent mini-challenge, review peer work and revise the monthly product.	Submit before/after evidence, a learning reflection and one specific next-step improvement.

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Week 4: Integrate, assess & reflect — testing and deployment planning

Session	Learning	Practical activity	Learner responsibility
1. Concept and inquiry	Explain API testing and deployment planning; define key terms, real-world use cases, common misconceptions and safe practice.	Investigate a familiar example of API testing and deployment planning; draw a diagram, annotate a worked example and discuss trade-offs.	Write a one-page concept note with two examples and three questions about testing and deployment planning.
2. Mentor demonstration	Observe an instructor-led demonstration of testing and deployment planning; connect decisions to the monthly product brief.	Follow a short worked example, then modify at least two elements without copying instructions.	Recreate the technique independently and explain how it contributes to full-stack community platform.
3. Guided studio and challenge	Apply testing and deployment planning in a realistic build; test assumptions, handle errors and use peer feedback.	Implement a small feature or artefact for full-stack community platform; test a normal case and an edge case.	Submit working evidence, a test checklist and a short explanation of design or code choices.
4. Independent build, review and reflection	Synthesise testing and deployment planning; diagnose problems, critique quality and document improvements.	Complete an independent mini-challenge, review peer work and revise the monthly product.	Submit before/after evidence, a learning reflection and one specific next-step improvement.

FairChild Institute

Month 10

Professional Software Engineering

Duration	4 weeks

Month Journey

Week 1: Discover & understand — Git and collaboration

Session	Learning	Practical activity	Learner responsibility
1. Concept and inquiry	Explain Git and collaboration; define key terms, real-world use cases, common misconceptions and safe practice.	Investigate a familiar example of Git and collaboration; draw a diagram, annotate a worked example and discuss trade-offs.	Write a one-page concept note with two examples and three questions about Git and collaboration.
2. Mentor demonstration	Observe an instructor-led demonstration of Git and collaboration; connect decisions to the monthly product brief.	Follow a short worked example, then modify at least two elements without copying instructions.	Recreate the technique independently and explain how it contributes to team-developed application.
3. Guided studio and challenge	Apply Git and collaboration in a realistic build; test assumptions, handle errors and use peer feedback.	Implement a small feature or artefact for team-developed application; test a normal case and an edge case.	Submit working evidence, a test checklist and a short explanation of design or code choices.
4. Independent build, review and reflection	Synthesise Git and collaboration; diagnose problems, critique quality and document improvements.	Complete an independent mini-challenge, review peer work and revise the monthly product.	Submit before/after evidence, a learning reflection and one specific next-step improvement.

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Week 2: Build core skills — testing and quality

Session	Learning	Practical activity	Learner responsibility
1. Concept and inquiry	Explain testing and quality; define key terms, real-world use cases, common misconceptions and safe practice.	Investigate a familiar example of testing and quality; draw a diagram, annotate a worked example and discuss trade-offs.	Write a one-page concept note with two examples and three questions about testing and quality.
2. Mentor demonstration	Observe an instructor-led demonstration of testing and quality; connect decisions to the monthly product brief.	Follow a short worked example, then modify at least two elements without copying instructions.	Recreate the technique independently and explain how it contributes to team-developed application.
3. Guided studio and challenge	Apply testing and quality in a realistic build; test assumptions, handle errors and use peer feedback.	Implement a small feature or artefact for team-developed application; test a normal case and an edge case.	Submit working evidence, a test checklist and a short explanation of design or code choices.
4. Independent build, review and reflection	Synthesise testing and quality; diagnose problems, critique quality and document improvements.	Complete an independent mini-challenge, review peer work and revise the monthly product.	Submit before/after evidence, a learning reflection and one specific next-step improvement.

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Week 3: Apply & challenge — code review

Session	Learning	Practical activity	Learner responsibility
1. Concept and inquiry	Explain code review; define key terms, real-world use cases, common misconceptions and safe practice.	Investigate a familiar example of code review; draw a diagram, annotate a worked example and discuss trade-offs.	Write a one-page concept note with two examples and three questions about code review.
2. Mentor demonstration	Observe an instructor-led demonstration of code review; connect decisions to the monthly product brief.	Follow a short worked example, then modify at least two elements without copying instructions.	Recreate the technique independently and explain how it contributes to team-developed application.
3. Guided studio and challenge	Apply code review in a realistic build; test assumptions, handle errors and use peer feedback.	Implement a small feature or artefact for team-developed application; test a normal case and an edge case.	Submit working evidence, a test checklist and a short explanation of design or code choices.
4. Independent build, review and reflection	Synthesise code review; diagnose problems, critique quality and document improvements.	Complete an independent mini-challenge, review peer work and revise the monthly product.	Submit before/after evidence, a learning reflection and one specific next-step improvement.

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Week 4: Integrate, assess & reflect — documentation and delivery

Session	Learning	Practical activity	Learner responsibility
1. Concept and inquiry	Explain documentation and delivery; define key terms, real-world use cases, common misconceptions and safe practice.	Investigate a familiar example of documentation and delivery; draw a diagram, annotate a worked example and discuss trade-offs.	Write a one-page concept note with two examples and three questions about documentation and delivery.
2. Mentor demonstration	Observe an instructor-led demonstration of documentation and delivery; connect decisions to the monthly product brief.	Follow a short worked example, then modify at least two elements without copying instructions.	Recreate the technique independently and explain how it contributes to team-developed application.
3. Guided studio and challenge	Apply documentation and delivery in a realistic build; test assumptions, handle errors and use peer feedback.	Implement a small feature or artefact for team-developed application; test a normal case and an edge case.	Submit working evidence, a test checklist and a short explanation of design or code choices.
4. Independent build, review and reflection	Synthesise documentation and delivery; diagnose problems, critique quality and document improvements.	Complete an independent mini-challenge, review peer work and revise the monthly product.	Submit before/after evidence, a learning reflection and one specific next-step improvement.

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Month 11

Security, Systems and AI-Native Engineering

Duration	4 weeks

Month Journey

Week 1: Discover & understand — security fundamentals

Session	Learning	Practical activity	Learner responsibility
1. Concept and inquiry	Explain security fundamentals; define key terms, real-world use cases, common misconceptions and safe practice.	Investigate a familiar example of security fundamentals; draw a diagram, annotate a worked example and discuss trade-offs.	Write a one-page concept note with two examples and three questions about security fundamentals.
2. Mentor demonstration	Observe an instructor-led demonstration of security fundamentals; connect decisions to the monthly product brief.	Follow a short worked example, then modify at least two elements without copying instructions.	Recreate the technique independently and explain how it contributes to security-reviewed AI-assisted prototype.
3. Guided studio and challenge	Apply security fundamentals in a realistic build; test assumptions, handle errors and use peer feedback.	Implement a small feature or artefact for security-reviewed AI-assisted prototype; test a normal case and an edge case.	Submit working evidence, a test checklist and a short explanation of design or code choices.
4. Independent build, review and reflection	Synthesise security fundamentals; diagnose problems, critique quality and document improvements.	Complete an independent mini-challenge, review peer work and revise the monthly product.	Submit before/after evidence, a learning reflection and one specific next-step improvement.

Week 2: Build core skills — privacy and threat thinking

Session	Learning	Practical activity	Learner responsibility
1. Concept and inquiry	Explain privacy and threat thinking; define key terms, real-world use cases, common misconceptions and safe practice.	Investigate a familiar example of privacy and threat thinking; draw a diagram, annotate a worked example and discuss trade-offs.	Write a one-page concept note with two examples and three questions about privacy and threat thinking.
2. Mentor demonstration	Observe an instructor-led demonstration of privacy and threat thinking; connect decisions to the monthly product brief.	Follow a short worked example, then modify at least two elements without copying instructions.	Recreate the technique independently and explain how it contributes to security-reviewed AI-assisted prototype.
3. Guided studio and challenge	Apply privacy and threat thinking in a realistic build; test assumptions, handle errors and use peer feedback.	Implement a small feature or artefact for security-reviewed AI-assisted prototype; test a normal case and an edge case.	Submit working evidence, a test checklist and a short explanation of design or code choices.
4. Independent build, review and reflection	Synthesise privacy and threat thinking; diagnose problems, critique quality and document improvements.	Complete an independent mini-challenge, review peer work and revise the monthly product.	Submit before/after evidence, a learning reflection and one specific next-step improvement.

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Week 3: Apply & challenge — AI-assisted engineering

Session	Learning	Practical activity	Learner responsibility
1. Concept and inquiry	Explain AI-assisted engineering; define key terms, real-world use cases, common misconceptions and safe practice.	Investigate a familiar example of AI-assisted engineering; draw a diagram, annotate a worked example and discuss trade-offs.	Write a one-page concept note with two examples and three questions about AI-assisted engineering.
2. Mentor demonstration	Observe an instructor-led demonstration of AI-assisted engineering; connect decisions to the monthly product brief.	Follow a short worked example, then modify at least two elements without copying instructions.	Recreate the technique independently and explain how it contributes to security-reviewed AI-assisted prototype.
3. Guided studio and challenge	Apply AI-assisted engineering in a realistic build; test assumptions, handle errors and use peer feedback.	Implement a small feature or artefact for security-reviewed AI-assisted prototype; test a normal case and an edge case.	Submit working evidence, a test checklist and a short explanation of design or code choices.
4. Independent build, review and reflection	Synthesise AI-assisted engineering; diagnose problems, critique quality and document improvements.	Complete an independent mini-challenge, review peer work and revise the monthly product.	Submit before/after evidence, a learning reflection and one specific next-step improvement.

Week 4: Integrate, assess & reflect — reliability and observability

Session	Learning	Practical activity	Learner responsibility
1. Concept and inquiry	Explain reliability and observability; define key terms, real-world use cases, common misconceptions and safe practice.	Investigate a familiar example of reliability and observability; draw a diagram, annotate a worked example and discuss trade-offs.	Write a one-page concept note with two examples and three questions about reliability and observability.
2. Mentor demonstration	Observe an instructor-led demonstration of reliability and observability; connect decisions to the monthly product brief.	Follow a short worked example, then modify at least two elements without copying instructions.	Recreate the technique independently and explain how it contributes to security-reviewed AI-assisted prototype.
3. Guided studio and challenge	Apply reliability and observability in a realistic build; test assumptions, handle errors and use peer feedback.	Implement a small feature or artefact for security-reviewed AI-assisted prototype; test a normal case and an edge case.	Submit working evidence, a test checklist and a short explanation of design or code choices.
4. Independent build, review and reflection	Synthesise reliability and observability; diagnose problems, critique quality and document improvements.	Complete an independent mini-challenge, review peer work and revise the monthly product.	Submit before/after evidence, a learning reflection and one specific next-step improvement.

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Month 12

Independent Engineering and Capstone

Duration 4 weeks

Month Journey

Week 1: Discover & understand — problem discovery

Session	Learning	Practical activity	Learner responsibility
1. Concept and inquiry	Explain problem discovery; define key terms, real-world use cases, common misconceptions and safe practice.	Investigate a familiar example of problem discovery; draw a diagram, annotate a worked example and discuss trade-offs.	Write a one-page concept note with two examples and three questions about problem discovery.
2. Mentor demonstration	Observe an instructor-led demonstration of problem discovery; connect decisions to the monthly product brief.	Follow a short worked example, then modify at least two elements without copying instructions.	Recreate the technique independently and explain how it contributes to year 1 integrated capstone.
3. Guided studio and challenge	Apply problem discovery in a realistic build; test assumptions, handle errors and use peer feedback.	Implement a small feature or artefact for year 1 integrated capstone; test a normal case and an edge case.	Submit working evidence, a test checklist and a short explanation of design or code choices.
4. Independent build, review and reflection	Synthesise problem discovery; diagnose problems, critique quality and document improvements.	Complete an independent mini-challenge, review peer work and revise the monthly product.	Submit before/after evidence, a learning reflection and one specific next-step improvement.

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Week 2: Build core skills — solution architecture

Session	Learning	Practical activity	Learner responsibility
1. Concept and inquiry	Explain solution architecture; define key terms, real-world use cases, common misconceptions and safe practice.	Investigate a familiar example of solution architecture; draw a diagram, annotate a worked example and discuss trade-offs.	Write a one-page concept note with two examples and three questions about solution architecture.
2. Mentor demonstration	Observe an instructor-led demonstration of solution architecture; connect decisions to the monthly product brief.	Follow a short worked example, then modify at least two elements without copying instructions.	Recreate the technique independently and explain how it contributes to year 1 integrated capstone.
3. Guided studio and challenge	Apply solution architecture in a realistic build; test assumptions, handle errors and use peer feedback.	Implement a small feature or artefact for year 1 integrated capstone; test a normal case and an edge case.	Submit working evidence, a test checklist and a short explanation of design or code choices.
4. Independent build, review and reflection	Synthesise solution architecture; diagnose problems, critique quality and document improvements.	Complete an independent mini-challenge, review peer work and revise the monthly product.	Submit before/after evidence, a learning reflection and one specific next-step improvement.

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Week 3: Apply & challenge — build and validate

Session	Learning	Practical activity	Learner responsibility
1. Concept and inquiry	Explain build and validate; define key terms, real-world use cases, common misconceptions and safe practice.	Investigate a familiar example of build and validate; draw a diagram, annotate a worked example and discuss trade-offs.	Write a one-page concept note with two examples and three questions about build and validate.
2. Mentor demonstration	Observe an instructor-led demonstration of build and validate; connect decisions to the monthly product brief.	Follow a short worked example, then modify at least two elements without copying instructions.	Recreate the technique independently and explain how it contributes to year 1 integrated capstone.
3. Guided studio and challenge	Apply build and validate in a realistic build; test assumptions, handle errors and use peer feedback.	Implement a small feature or artefact for year 1 integrated capstone; test a normal case and an edge case.	Submit working evidence, a test checklist and a short explanation of design or code choices.
4. Independent build, review and reflection	Synthesise build and validate; diagnose problems, critique quality and document improvements.	Complete an independent mini-challenge, review peer work and revise the monthly product.	Submit before/after evidence, a learning reflection and one specific next-step improvement.

Week 4: Integrate, assess & reflect — showcase and reflection

Session	Learning	Practical activity	Learner responsibility
1. Concept and inquiry	Explain showcase and reflection; define key terms, real-world use cases, common misconceptions and safe practice.	Investigate a familiar example of showcase and reflection; draw a diagram, annotate a worked example and discuss trade-offs.	Write a one-page concept note with two examples and three questions about showcase and reflection.
2. Mentor demonstration	Observe an instructor-led demonstration of showcase and reflection; connect decisions to the monthly product brief.	Follow a short worked example, then modify at least two elements without copying instructions.	Recreate the technique independently and explain how it contributes to year 1 integrated capstone.
3. Guided studio and challenge	Apply showcase and reflection in a realistic build; test assumptions, handle errors and use peer feedback.	Implement a small feature or artefact for year 1 integrated capstone; test a normal case and an edge case.	Submit working evidence, a test checklist and a short explanation of design or code choices.
4. Independent build, review and reflection	Synthesise showcase and reflection; diagnose problems, critique quality and document improvements.	Complete an independent mini-challenge, review peer work and revise the monthly product.	Submit before/after evidence, a learning reflection and one specific next-step improvement.